

A Project on

**SOFT ACTOR CRITIC BASED SEMANTIC AWARE RESOURCE ALLOCATION FOR HYBRID
SEMANTIC/BIT NOMA NETWORKS**

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
25SC2107E**

in

A.Y.: 2026-27

by

Name N. Jesly Monica

roll 2520040052

Name M.Shreya

roll 2520040063

Name P.Karthika

roll 2520040094

Under the Esteemed guidance of

Mr. N Lava Prasad

Assistant Professor,

Department of ECE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Name of Title: Soft actor critic based semantic aware resource allocation for hybrid semantic/bit NOMA Networks

1. Abstract:

The increasing demand for wireless services with different communication requirements has created the need for resource allocation techniques that can efficiently support both conventional data transmission and emerging semantic communication. In a Hybrid Semantic/Bit Non-Orthogonal Multiple Access (NOMA) network, semantic users and conventional bit users share the available wireless resources, making resource allocation more challenging due to differences in channel conditions, transmission requirements, and quality of service. Conventional resource allocation methods mainly focus on parameters such as data rate, transmit power, and spectral efficiency, but they may not sufficiently consider the importance and quality of the information being transmitted by semantic users. In addition, the joint allocation of resources among heterogeneous users becomes difficult when channel conditions and user requirements change dynamically. To address these challenges, this work proposes a Soft Actor-Critic (SAC) based semantic-aware resource allocation framework for Hybrid Semantic/Bit NOMA networks. The proposed system models resource allocation as a continuous-control reinforcement learning problem, where the SAC agent interacts with the wireless network environment and learns an optimal resource allocation policy. The state observed by the agent includes relevant network information such as channel conditions, user requirements, and available transmission resources. Based on the observed state, the SAC agent determines suitable resource allocation actions, particularly transmit power allocation among semantic and conventional bit users. The reward function is designed to encourage efficient resource utilization while maintaining the required communication performance of both types of users. Through repeated interaction with the network environment, the SAC agent learns how to adapt resource allocation according to changing network conditions and user requirements.

The proposed framework will be implemented using Python-based simulations and evaluated under different network conditions, numbers of users, channel conditions, and proportions of semantic and conventional bit users. The performance of the SAC-based resource allocation strategy will be compared with conventional resource allocation approaches to determine its effectiveness in managing the limited network resources. The proposed system will be evaluated using parameters such as system sum rate, spectral efficiency, energy efficiency, transmit power consumption, achievable data rate, and user fairness. The quality of service provided to semantic and conventional bit users will also be analyzed to determine whether the learned allocation policy can maintain a suitable balance between the two communication modes. In addition, reinforcement learning performance will be evaluated using cumulative reward, convergence behavior, and training stability. The impact of different network conditions on the learned resource allocation policy will also be studied to examine its adaptability. Through the use of Soft Actor-Critic, the proposed work aims to develop an adaptive resource allocation mechanism that can intelligently distribute wireless resources between semantic and bit-based users while improving overall network efficiency and maintaining the communication requirements of heterogeneous users. The results are expected to demonstrate the suitability of deep reinforcement learning for semantic-aware resource management in Hybrid Semantic/Bit NOMA networks.

2. SOFTWARE REQUIREMENTS:

s.No	Software	Purpose
1	Python 3.x	Main programming language for the complete project
2	Google Colab	Cloud-based environment for writing and running Python code
3	Jupyter Notebook / JupyterLab	Interactive development, experiments, and result analysis

4	PyTorch	Implementing the SAC Actor, Critic, target networks, and neural-network training
5	Gymnasium	Creating the custom NOMA reinforcement-learning environment
6	NumPy	Numerical calculations, matrices, channel coefficients, and resource allocation calculations
7	Pandas	Storing, organizing, and analyzing simulation results
8	Matplotlib	Generating performance graphs and comparison plots
9	SciPy	Mathematical, statistical, and scientific computations
10	Scikit-learn	Data preprocessing, normalization, and optional baseline methods

Signature of the Supervisor

Course Instructor